

(ii) isolating the compound of Formula (I) obtained in step (i) from the amount of the compound of Formula (VI);

[0037] In embodiments, the process further comprises:

(iii) optionally adding about 1 additional equivalent of the compound of Formula (VI) to the remaining amount of the compound of Formula (VI) from step (ii);

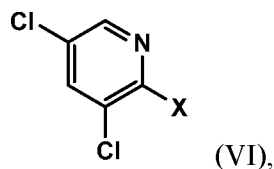
(iv) repeating step (i) using the compound of Formula (VI) in step (ii) or step (iii) and about 1 equivalent of the activator; and

(v) optionally repeating steps (ii) to (iv).

[0038] In embodiments, step (iii) comprises in total at least about 3 equivalents of the compound of Formula (VI).

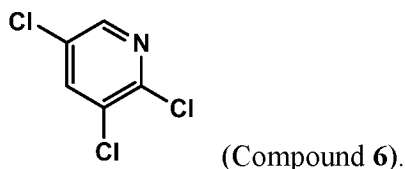
[0039] In embodiments, the process comprises about 2-8 total cycles to form the compound of Formula (I).

[0040] In embodiments, a compound of Formula (I-F) is prepared by a process comprising a step (i-F), wherein said step comprises contacting a compound of Formula (VI) or a salt thereof



with a fluoride source, and wherein X is a halogen.

[0041] In embodiments, a compound of Formula (VI) is



[0042] In embodiments, the fluoride source is a fluoride salt, e.g., NaF, LiF, RbF, CsF, and KF.

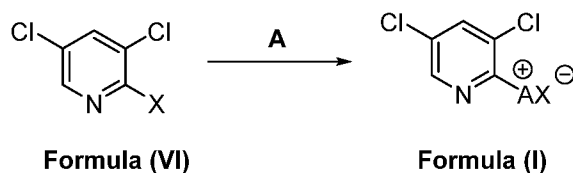
[0043] In embodiments, the fluoride source is KF.

[0044] In embodiments, the KF is spray-dried KF.

[0045] In embodiments, step (i-F) further comprises a solvent.

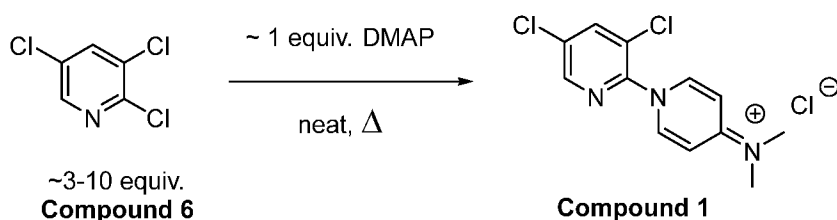
**Scheme 1. Exemplary Step (i)**

a.



**A** = Nucleophile Activator such as DMAP, DABCO, pyridine, and the like  
**X** = halogen

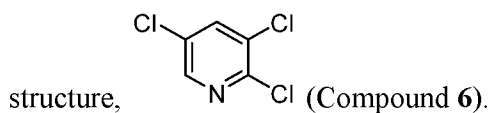
b.



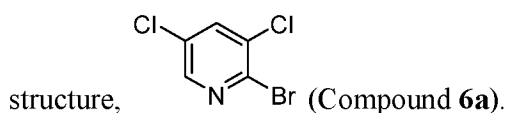
**[0106]** As shown in **Scheme 1a**, a step (i) can comprise contacting a trihalogenated pyridine compound of Formula (VI) with a nucleophile activator **A** to form a compound of Formula (I). A further exemplary embodiment of a step (i) is shown in **Scheme 1b**.

**[0107]** In embodiments, **X** is F, Cl or Br.

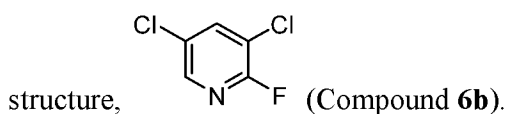
**[0108]** In embodiments, **X** is Cl, and the compound of Formula (VI) has the following



**[0109]** In embodiments, **X** is Br, and the compound of Formula (VI) has the following



**[0110]** In embodiments, **X** is F, and the compound of Formula (VI) has the following



**[0111]** In embodiments, **A** is a nucleophilic amine, a triarylphosphine, or a trialkylphosphine. In embodiments, **A** is DMAP (4-dimethylaminopyridine), DABCO (1,4-diazabicyclo[2.2.2]octane), DBU (1,8-Diazabicyclo[5.4.0]undec-7-ene), DBN (1,5-